Example 6.5a: #Program to illustrate the use of 'in' and 'not in' in if statement

```
ch=input ("Enter a character :")
# to check if the letter is vowel
if ch in ('a', 'A', 'e', 'E', 'i', 'I', 'o', 'O', 'u', 'U'):
        print (ch,' is a vowel')
# to check if the letter typed is not 'a' or 'b' or 'c'
if ch not in ('a', 'b', 'c'):
        print (ch,'  the letter is not a/b/c')
```

Output 1:

```
Enter a character :e
e is a vowel
```

Output 2:

```
Enter a character :x
x the letter is not a/b/c
```

6.2.3. Iteration or Looping constructs

Iteration or loop are used in situation when the user need to execute a block of code several of times or till the condition is satisfied. A **loop** statement allows to execute a statement or group of statements multiple times.

BX6SWB

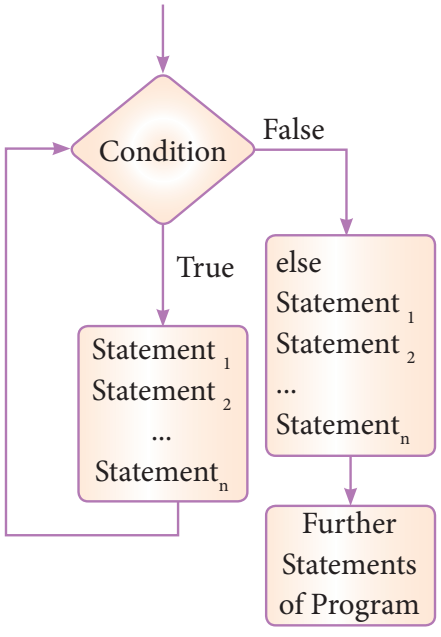

Fig 6.3 Diagram to illustrate how looping construct gets executed

Python provides two types of looping constructs:

- while loop
- for loop